Basic Electronics

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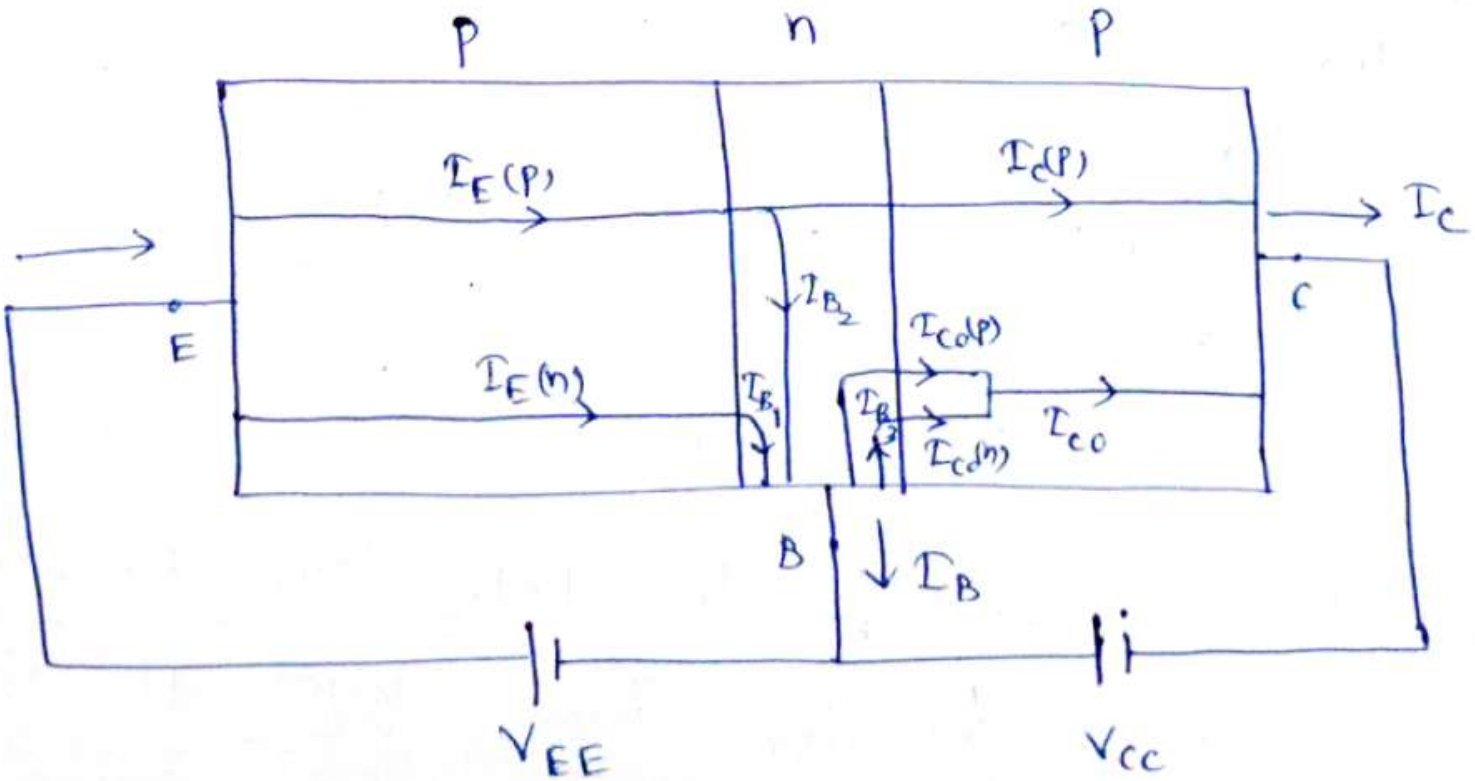
Current components



$$I_E = I_E(p) + I_E(n)$$

$$I_C = I_C(p) + I_{Co}$$

$I_E = I_C + I_B$

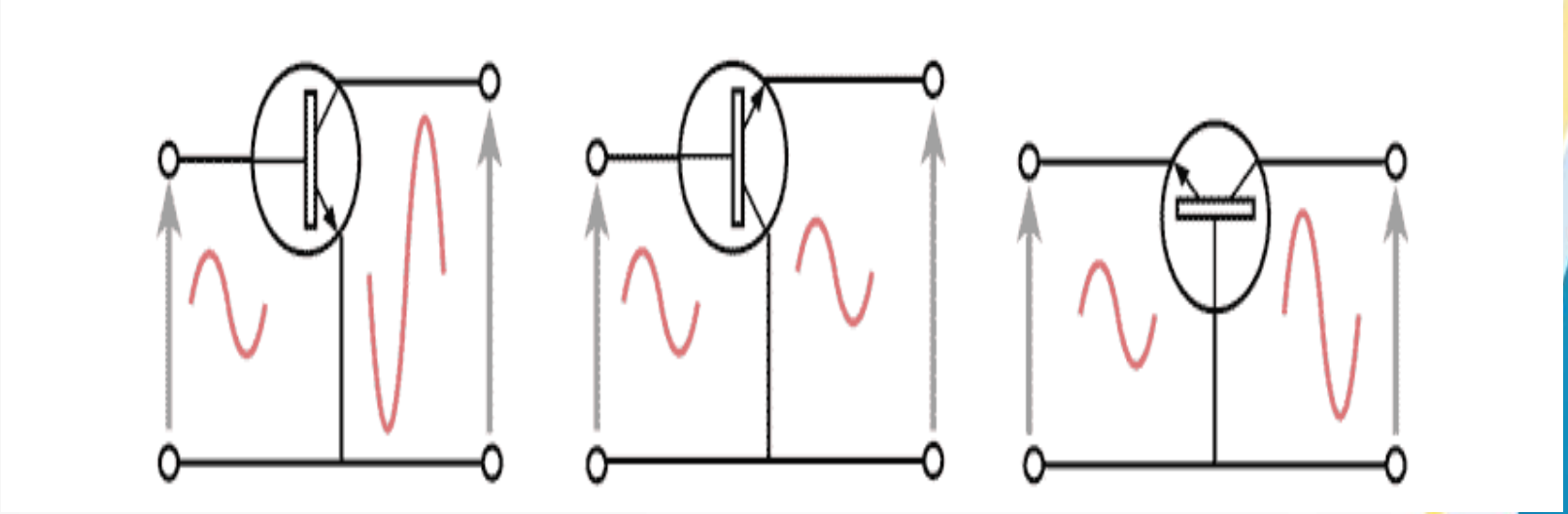
$I_C = \alpha I_E$

Contd...

- Trannsistor amplyfing action is due to it's capacity of transfer it's signal current from low resistance path to high resistance path.
- TRANSFER+RESISTOR=TRANSISTOR
- <u>Configuration of Transistors</u>
- As the Bipolar Transistor is a three terminal device, there are basically three possible ways to connect it within an electronic circuit with one terminal being common to both the input and output.
- Each method of connection responding differently to its input signal within a circuit as the static characteristics of the transistor vary with each circuit arrangement.

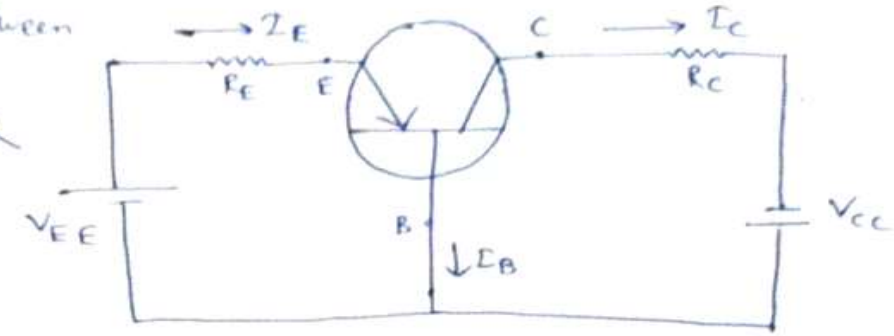
Configuration of Transistors

- 1. Common Base Configuration (CB) - has Voltage Gain but no Current Gain.
- 2 Common Emitter Configuration (CE) - has both Current and Voltage Gain.
- 3. Common Collector Configuration (CC) - has Current Gain but no Voltage Gain.



CB Confuguration

- Here i/p is applied between Emitter and Base while o/p is taken between Collector and Base



- So Base is common to both i/p and o/p and named as CB Config

Current amplification factor (α)

- when there is no i/p a.c signal applied, the ratio of o/p collector current to the i/p emitter current is called as dc alpha (α_{dc}) of transistor.

$$\boxed{\alpha_{dc} = \frac{I_C}{I_E}}$$

α measures the quality of transistor.

α lies between (0.96–0.995)

Contd...

$$\Rightarrow \quad I_C = \alpha I_E$$

we know $\quad I_E = I_C + I_B$

$$\Rightarrow \quad I_E = \alpha I_E + I_B$$

$$\Rightarrow \quad \boxed{I_B = (1-\alpha) I_E}$$

If we consider leakage current

$$I_C = \alpha I_E + I_{CBO} = \alpha (I_C + I_B) + I_{CBO}$$

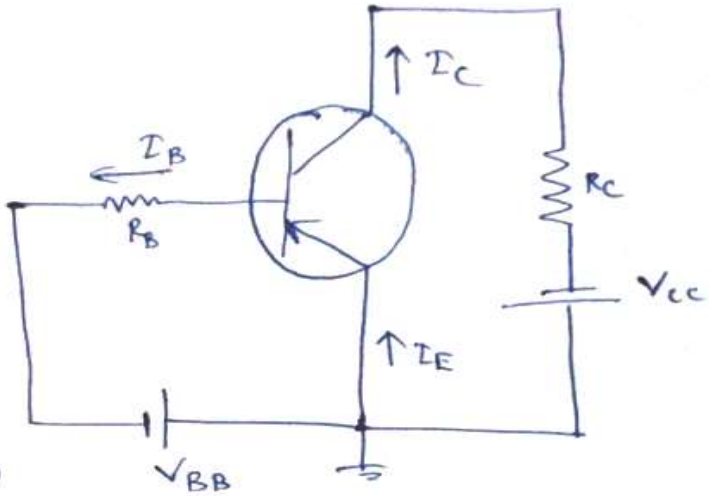
$$I_C - \alpha I_C = \alpha I_B + I_{CBO}$$

$$\Rightarrow \boxed{I_C = \frac{\alpha}{1-\alpha} I_B + \frac{1}{1-\alpha} I_{CBO}}$$

($I_{CO} = I_{CBO}$ for CB configuration)

CE Configuration

- Here i/p is applied between Emitter and base while o/p is taken between collector and emitter.



- So emitter is common to both i/p and o/p and named as CE configuration

Current Amplification factor (β)

- when no ac signal is applied, then the ratio of collector current (o/p) to base current (i/p) is called as β_{dc}.

$$\boxed{\beta = \frac{I_C}{I_B}}$$

β lies between 25-300

$$I_C = \beta I_B$$

Contd...

If leakage current considered then $I_C = \beta I_B + I_{CEO}$

$$I_C = \beta I_B + \frac{1}{1-\alpha} I_{CBO}$$

$I_{CEO} \longrightarrow$ Leakage current in CE configuration

$I_{CBO} \longrightarrow$ leakage current in CB configuration

$$\boxed{I_{CEO} = \frac{1}{1-\alpha} I_{CBO}}$$

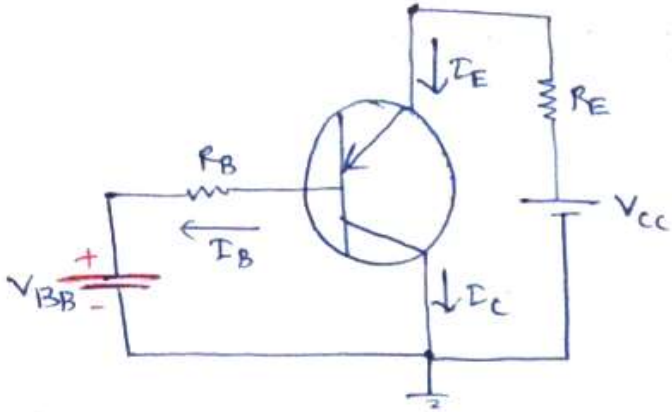
CC configuration :-

- Here i/p is applied between collector and Base while o/p is taken between emitter and collector.
- So collector is common between both i/p and o/p & hand as CC configuration.

CC Configuration

Current Amplification factor (γ)

- when no ac signal applied, the ratio of emitter current to base current is called γ_{dc}.



$$\boxed{\gamma = \frac{I_E}{I_B}}$$

we know $I_E = I_C + I_B$

$$I_C = \alpha I_E + I_{CBO}$$

$$\Rightarrow I_E = \alpha I_E + I_{CBO} + I_B$$

$$\Rightarrow I_E(1-\alpha) = I_B + I_{CBO} \Rightarrow I_E = \frac{1}{1-\alpha} I_B + \frac{1}{1-\alpha} I_{CBO}$$

$$\Rightarrow \boxed{I_E = (1+\beta) I_B + (1+\beta) I_{CBO}}$$

Relation Between α, β, γ :-

$$\alpha = \frac{I_C}{I_E}, \quad \beta = \frac{I_C}{I_B}, \quad I_E = I_C + I_B \Rightarrow I_B = I_E - I_C$$

$$\beta = \frac{I_C}{I_B} = \frac{I_C}{I_E - I_C} = \frac{I_C / I_E}{1 - \frac{I_C}{I_E}} = \boxed{\frac{\alpha}{1-\alpha} = \beta}$$

$$\Rightarrow \beta = \frac{\alpha}{1-\alpha} \Rightarrow \beta(1-\alpha) = \alpha$$

$$\Rightarrow \beta - \alpha\beta = \alpha$$

$$\Rightarrow \alpha(1+\beta) = \beta \Rightarrow \boxed{\alpha = \frac{\beta}{1+\beta}}$$

$$\gamma = \frac{I_E}{I_B}, \quad \alpha = \frac{I_C}{I_E} \Rightarrow I_B = I_E - I_C$$

$$\gamma = \frac{I_E}{I_B} = \frac{I_E}{I_E - I_C} = \frac{1}{1 - \frac{I_C}{I_E}} = \boxed{\frac{1}{1-\alpha} = \gamma}$$

$$\Rightarrow \gamma(1-\alpha) = 1 \Rightarrow \gamma - \gamma\alpha = 1 \Rightarrow \boxed{\alpha = \frac{\gamma - 1}{\gamma}}$$

$$\boxed{\alpha = \frac{\beta}{1+\beta} = \frac{\gamma - 1}{\gamma}}$$